

Effects of root systems on crack formation: experiments, modeling, and analyses

Rui Nong^{a,b}, Yukuai Wan^{a,b}, Yimin Ding^{a,b}, Mingtang Chai^{a,b}, Lei Zhu^{a,b,*}

^a School of Civil and Hydraulic Engineering, Ningxia University, Yinchuan, Ningxia 750021, China

^b Key Laboratory of the Internet of Water and Digital Water Governance of the Yellow River, Ningxia University, Yinchuan 750021, China

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ABSTRACT

Plant roots can increase soil shearing resistance directly by mechanical reinforcing. The 2D crack pattern of the surface soil near the plant were observed experimentally, and it was found that the soil cracking near the plant roots was inhibited, which may be due to the influence of the plant roots. This paper focuses on the mechanical reinforcement of soil by plant roots and investigates the development rule and formation mechanism of desiccation cracking under the influence of plant roots. A numerical model describing the process of desiccation cracking of soils in the presence of root systems, which is based on a lattice of Hooke springs model, was combined with a non-stationary random field characterized by a trend term and a random fluctuation term. The root density distribution function was introduced to describe the root distribution, and an exponential function of critical strain and root density was constructed to obtain the trend of critical strain distribution of the root systems (the root systems of a single plant). The random fluctuation term was generated by the conditional simulation technique. Minkowski densities (i.e., area density, length density and Euler number density) were adopted to quantify the crack patterns. Parameters of the model were calibrated and verified by the field experimental results, and its precision was evaluated. The effect of most input parameters (include the new parameters maximum critical strain value h_c and the lateral distribution radius of RDDF r_m) on the model output was analyzed. The results show that in the validation process the coefficients of determination R^2 for the Minkowski densities is between 0.749 and 0.948. Root mean square error is between 0.000 2–0.120 9. *BIAS* is between 0.049 5–0.139 0, and the consistency index is between 0.908 and 0.987. The model is effective in predicting the formation and development of cracks in farmland soils under the influence of the root systems. The model produces the dynamic development of the desiccation cracking network of farmland soils under the influence of the root systems. The simulation results provide ample indication of the reinforcing effect of the root systems on the soil and the limiting effect of the root systems on desiccation cracking.

1. Introduction

Desiccation cracking of soils is a common and natural phenomenon (Tran et al., 2019). Crack networks provide preferential flow paths for water and solute transport, which greatly increases the hydraulic conductivity of the soil (Narani et al., 2020; Wei et al., 2020; Zhu et al., 2018). However, it also leads to problems such as waste of irrigation water (Bronswijk, 1991; Dekker, Bouma, 1984), erosion of deep soil layer (Baker, 1981; Tang et al., 2012), and leakage of landfill gas and leachate (Flury et al., 1994; Kurtzman et al., 2016). Besides, Desiccation cracking induces salinization of the soil by accelerating deep evaporation. (Baram et al., 2013; Kurtzman et al., 2016). Mechanical responses

of soils subjected to desiccation cracking can also be severely affected, resulting in impaired strength, excessive deformation, and increased compressibility (Tang et al., 2007; Tang et al., 2018). These effects lead to slope problems (Li et al., 2020; Stirling et al., 2020; Zhang et al., 2020), embankment damage (Foster et al., 2000), foundation failure (Lozada et al., 2015), and liner failure (Li et al. (2016); Atique, Sanchez (2011)). Desiccation cracking will threaten the long-term performance of materials, such as leakage from earth-rock dams and liners. Therefore, preventing or reducing the occurrence of desiccation cracking can increase the service life of soil materials, thereby mitigating or overcoming the damages caused by desiccation cracking.

Root-water uptake affects the moisture dynamics and redistribution

* Corresponding author at: School of Civil and Hydraulic Engineering, Ningxia University, Yinchuan, Ningxia 750021, China.

E-mail address: nxuzhulei@163.com (L. Zhu).

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of suction in the vadose zone (Ni et al., 2018; Bordoloi et al., 2020). Induced soil suction in vegetated soil can be several times higher than that in bare ground (Garg, 2015; Ng et al., 2016a, 2016b). On the other hand, plant roots have been reported to exert an active effect in restricting desiccation cracking of soils (Li et al., 2016). It can act as a soil reinforcement, providing additional shear strength to soils (Wu and Watson, 1998; Roering et al., 2003). This provides a feasible method for preventing desiccation cracking. Studying the effects of plant roots on desiccation cracking of soils can help to improve the understanding of how the root systems restricts desiccation cracking, and can further promote related technological and engineering innovation. Therefore, the influence of plant roots on desiccation cracking has been of considerable interest to scholars in agriculture, geology, and environmental disciplines. The effect of root systems on the development of soil cracks is explored in this paper with regard to the characteristics of root reinforcement.

There has been considerable research investigating the effect of plant roots or fiber materials as soil additives on the development of cracks. It is believed that plant roots and fiber materials have similar mechanisms of reinforcement of the soil. Plant roots as soil reinforcement can increase soil strength and resist soil cracking by performing like a fiber in soil (Zhou et al., 2009). Zhou and Wang (2019) studied the meso-mechanics characteristics of soil reinforcement by plant roots, and the results showed that plant roots inhibited further expansion of cracks and their bridging behavior contributed to the strength and toughness of the soil. Li et al. (2016) demonstrated that the root systems restricted the formation of cracks by image analysis in the experimental soil columns. Zhang et al. (2014) studied the desiccation cracking of soils with different contents of root systems and found that the higher the root content of the soil, the smaller the crack area. Zhang et al. (2019) used fractal theory to analyze the desiccation cracking morphology of farmland soil under the influence of root systems, and the study showed that root systems exerted an inhibitory effect on the desiccation cracking of farmland soils. Plant roots help to improve the tensile strength of the soil and have a certain inhibitory effect on cracking of soils. However, previous studies have focused on the inhibitory effect of plant roots as soil additives on cracking of soils, while there are few studies on the effect of root systems on the morphology of desiccation cracking of soils in the natural state.

The numerical computation method of soil cracking physical models have been developed into linear elastic fracture mechanics theory model, Finite Element Method (FEM) and The Discrete Element Method (DEM) (Aoki et al., 2002, 2007; Horgan and Young, 2000; Musielak and Śliwa, 2015), but studies on the influence of soil desiccation cracking models by root systems have not been reported. Horgan and Young (2000) presented an empirical stochastic model based on the geometry of two-dimensional crack growth, and the simulation results were compared with cracks in real soil. Konrad and Ayad (1997) constructed a predictive framework for crack spacing using linear elastic fracture mechanics theory to briefly describe the crack extension phenomenon. Unlike the model based on the theory of linear elastic fracture mechanics, the Maxwell model has been proposed to treat clay materials as viscoelastic bodies to simulate the cracking process (Aoki et al., 2002, 2007; Musielak and Śliwa, 2015), and good simulation results have been obtained. Finite Element Method (FEM) based on continuum computational mechanics has been applied to study desiccation cracking of soils (Bui et al., 2015; Youshida and Adachi, 2004). Sánchez et al. (2014) adopted the mesh fragmentation technique to standard finite element programs and analyzed multiple case studies related to desiccation cracking of soils developed under laboratory and field conditions. Vo et al. (2017) used FEM to investigate the crack initiation and propagation process, and considered the coupling between hydraulic and mechanical behavior. However, FEM simulations of the cracking process generally presuppose only a few cracks or uses model boundary as the potential crack, which inevitably undermines the capability of prediction on desiccation cracking initiation and propagation (Gui et al., 2016,

2018). The Discrete Element Method (DEM), which treats the soil as an assemblage of discrete elements, has good potential for simulating desiccation cracking (Vo et al., 2017). Peron et al. (2009) simulated shrinkage at the microscopic scale, as well as crack initiation, expansion and geometry using DEM. Combined with laboratory observations, Gui et al. (2016) used the hybrid continuum-discrete element method to simulate soil desiccation cracking, and identified that the shear strength, tensile strength, and soil sample thickness exerted a significant controlling effect on desiccation cracking. Although DEM is efficient in simulating desiccation cracking, it is unable to predict multi-physical processes (e.g., moisture evaporation, heat and mass exchanges) in soil desiccation cracking (Bui et al., 2015).

In a previous study, Vogel et al. (2005b) developed a physically-based model to simulate crack formation based on Hooke's law. Compared with the numerical model proposed by Konrad and Ayad (1997a), Vogel's model can reproduce the realistic overall appearance of the crack pattern and simulate the whole dynamic development of the cracking (Vogel et al., 2005b). Furthermore, it can capture prominent features of the nonlinear dynamics of crack network development observable in nature (i.e., the characteristic shape of aggregates and the characteristic angles of bifurcations). More specifically, Vogel's model was applied to simulate dynamic crack development caused by water evaporation, where heterogeneity was introduced through a critical threshold of strain for each spring connecting two nodes (Zhu et al., 2020). The critical strain for each spring was described by a Gaussian distribution, with nodes having mean and variance in space being identified. Zhu et al. (2020) introduced the correlation length in geostatistics to the Vogel et al. (2005b) model to reflect the spatial correlation of the critical strain.

Generally, stationary random fields and quasi-stationary random fields are adopted to describe the spatial distribution variability of critical soil strain. However, these model are hardly representative of the spatially varying trends in soil tensile strength caused by root anchorage. Non-stationary random fields are widely used in the geotechnical problem. In the characterization process of non-stationary random fields, the discrete variables of soil parameters can generally be divided into a trend function and a fluctuating component (Shu et al., 2020). In farmland, the tensile strength of soil under the influence of root systems exhibits significant spatial variability, and the tensile strength of soil is related to root density. In this paper, the empirical root density distribution function proposed by Vrugt et al. (2001) was introduced to describe the root distribution (axisymmetric distribution along the center of a single plant), and an exponential relationship between root density and critical strain was constructed. Then, the spatial distribution trend of the critical strain at the root systems was obtained. Although the critical strain of the soil could not be measured directly, the critical strain within the influence range of the root systems was significantly larger due to the reinforcement of the root systems. Therefore, the maximum critical strain value (the maximum value of the trend term), whose value should be significantly greater than the average value of the critical strain of the soil, at the center of plants and the lateral distribution radius of RDDF r_m could be assumed to be undetermined parameters, and the simulation results could then be compared with the experimental images to calibrate the parameters.

In the field experiments, the locations of the plant were known. Due to the root reinforcement, the critical strain in plant-root soil was larger than in bare soil. To consider these prior information, the conditional random field was taken as a random fluctuation term to describe the random perturbation of critical strain relative to its mean trend, which could reduce uncertainty over the length range of correlation. Also, the correlation of soil critical strain will be described in the conditional simulation. Therefore, further improvement of the model proposed by Vogel et al. (2005b) could be achieved.

The main contents of this paper can be seen as follows. Firstly, the force and energy calculation formulas of the Hookean spring model are introduced. Secondly, the process of soil cracking experiment under the

influence of root systems is described. Subsequently, the relationship between the model parameter values and crack patterns was demonstrated by Monte-Carlo simulation, and the optimized parameter set by minimizing the difference between the predicted and measured Minkowski numbers (quantification indicators of cracking, namely, area density, length density and Euler number density). Finally, the sensitivity of the simulation results to the maximum critical strain value h_c (represents the strength of root reinforcement to the soil) and the lateral distribution radius of RDDF r_m (represents the lateral extension of the root systems of individual plants) were analyzed.

2. The non-stationary desiccation cracking model

The desiccation cracking model of soils under the influence of the root systems proposed in this paper is an extension of the physically-based model for crack formation originally developed by Vogel et al. (2005b) and an improved model by Zhu et al. (2020). A non-stationary critical strain random field was generated using the exponential decreasing trend of the critical strain along the radial distance of the root systems as the trend term and the conditional stationary random field as the random fluctuation term. In addition to the mean, variance, and correlation length, the simulation of the non-stationary random field also depends on three new parameters, namely the number of the maximum critical strain locations n , the maximum critical strain value h_c and the lateral distribution radius of RDDF r_m . They represent the number of plants, the reinforcement strength of the soil by the root systems, the influence range of root systems to soil reinforcement, respectively. The physically-based model of Vogel et al. (2005b) was then used to describe the crack formation. The simulation attempts to reproduce the complex crack pattern of the soil under the influence of the root systems.

2.1. Vogel et al. (2005b) model

The model (Vogel et al., 2005b) constructed a two-dimensional flat layer where cracks form as a result of slow contraction of the material due to water evaporation. The soil layer was regarded as an elastic material and was represented by a triangular network of simple Hookean springs. Each node was connected to its six neighbors by a spring with constant K and natural length d (the length of the relaxed spring). In this paper, the same calculation as in Vogel et al. (2005b) was chosen and K was set to 1. The strain ε_{ij} of any two nodes, the net force F_i acting on node i through its n neighbors, and the total energy E_i at each node, can be simplified as (Vogel et al., 2005b; Zhu et al., 2020):

$$\varepsilon_{ij} = \frac{|\mathbf{x}_i - \mathbf{x}_j|}{d} - 1 \quad (1)$$

$$\mathbf{F}_i = - \sum_{j \in n} K_{ij} \frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|} [|\mathbf{x}_i - \mathbf{x}_j| - d] \quad (2)$$

$$E_i = \frac{1}{2} \sum_{j \in n} K_{ij} [|\mathbf{x}_i - \mathbf{x}_j| - d]^2 \quad (3)$$

where $\mathbf{x}_i, \mathbf{x}_j$ are the vector positions for any two nodes, $|\mathbf{x}_i - \mathbf{x}_j| = 1$ denotes the distance of the nodes in the initial lattice with arbitrary units. And d stands for the length of the relaxed spring, which is initially set to 1 so that all springs are completely relaxed. Then, desiccation was simulated by decreasing d , resulting in an increase in tension of the springs. During desiccation, the actual strain was compared to the critical threshold strain and if the strain between the two nodes exceeded the critical threshold, the spring would break. The breakage of the spring had immediate consequences for the forces on the nodes previously connected by that spring. They were pulled apart by the other springs, which were still connected. The Friction μ was one free parameter of the model. When friction $\mu < F$, the nodes moved towards a new location of

minimal energy, as can be seen in Eq. (3). The maximum number of iterations nit was a control parameter that was executed each time after a spring was broken. Expansion of the springs $u = 1 - d$ was used as a substitute for time. The boundary condition was specified so that the boundary nodes were not allowed to move perpendicular to the boundary.

2.2. Non-stationary random field of critical strain

According to Vogel et al. (2005b), heterogeneity was introduced through the critical strain threshold of each spring connecting two nodes. The threshold for each spring was taken from a Gaussian probability distribution with specified mean and variance and was distributed randomly as white noise in space, without regard to spatial correlation. Zhu et al. (2020) introduced the correlation length into the Vogel et al. (2005b) model to describe the spatial correlation of critical strain. However, the reinforcement effect of the root systems on the soil made the spatial variation structure of the critical strain more complicated. The mean critical strain of root-bearing soil showed a non-stationary distribution trend. We proposed a non-stationary random field model in which the critical strain at the plants tends to be exponentially distributed.

The spatial trend of critical strain was characterized by a trend term and a random fluctuation term. The two-dimensional non-stationary critical strain random field H_u under the influence of root systems can be expressed as:

$$H_u(\mathbf{x}) = \sum_{i=1}^n H_b(\mathbf{x}_i) + I_c(\mathbf{x}) \mathbf{x}_i \in V \quad (4)$$

where $H_b(\mathbf{x}_i)$ is a trend term, which is taken as the mean of critical strain along the corresponding radial distance at the plants. $I_c(\mathbf{x})$ stands for a random fluctuation term, which is applied to describe the random fluctuation of critical strain relative to its mean trend. V represents the vector space of the influence range of root systems.

In general, the tensile strength of soil at the plant increases with the increase of root density (Tengbeh, 1993) and De Baets et al. (2008) performed an exponential fitting of root density and soil cohesion. An exponential function was constructed to describe the relationship between critical strain and root density, and the trend term H_b could be expressed as follows:

$$H_b(\mathbf{x}_i) = e^{K_r B(\mathbf{x}_i)} \quad (5)$$

where $B(\mathbf{x}_i)$ characterizes the dimensionless spatial root distribution and K_r is an empirical parameter used to scale the relative root density distribution to absolute root density.

Due to the complexity of the soil-root systems, it is difficult to evaluate the RDDF through field and laboratory experiments (Feddes et al., 1978; Jarvis, 1989), so the RDDF is usually determined by empirical models. In order to describe the soil critical strain field at a two-dimensional scale, the radial relative root density distribution function proposed by Vrugt et al. (2001) was adopted:

$$B(\mathbf{x}_i) = \left[1 - \frac{|\mathbf{x}_i - \mathbf{x}_j|}{r_m} \right] e^{-\frac{P_r}{r_m} |r^* - |\mathbf{x}_i - \mathbf{x}_j||} \quad (6)$$

where $|\mathbf{x}_i - \mathbf{x}_j|$ represents the radial distance from the origin of the plant, r_m represents the maximum rooting length in the radial direction (L) and P_r and r^* are empirical parameters with units (–) and (L), respectively. This function reflects only the root density distribution in the plane and does not take into account the variation with depth. In this study, a root architecture from Vrugt (2001) was chosen, which assumes axial symmetry of horizontal root distribution and the empirical parameters r^* were set as 0, so that both, the maximum root density and the maximum critical strain were located in the center of the plants. It is assumed that the same plant roots have the same reinforcement effect on the soil. That

is, the parameters r_m , P_r and r^* for a certain plant are determined constants. The maximum critical strains $H_b(x_{i1})$, $H_b(x_{i2})$, ..., $H_b(x_{in})$ are equal, which are located at x_{i1} , x_{i2} , ..., x_{in} , and are represented by h_c . ' x_{i1} , x_{i2} , ..., x_{in} ' are the locations at the center of plants.

Conditional simulation refers to a random field simulation method that uses measurement data as constraints (Chen et al., 2002; Lu and Zhang, 2004; Shi et al., 2007, 2009). Conditional simulation has the following characteristics: i) the simulation value of the random field at the measurement point is equal to the measured value while keeping the overall spatial statistics (mean, variance and correlation) (Zha et al., 2017); ii) increasing the number of measured values can reduce the uncertainty of the random field simulation. It is expected that the true critical strain at the center of plants can be determined by the maximum critical strain h_c , that is, the random perturbation term at the center of plants is a constant. Therefore, the conditional random field was introduced as a random perturbation term I_c with mean $\bar{e}$ and variance δ^2 , and n 'virtual observations' $I(x_1)$, $I(x_2)$, ..., $I(x_n) = \bar{e}$ was set at the center of the plants as constraint.

Suppose we have n 'virtual observations' $I(x_i) = \bar{e}$, located at x_i ($i = 1, 2, \dots, n$), the conditional mean and covariance of I can be derived from the kriging technique (Lu and Zhang, 2004; Zhang, 2005). x is the position vector and $x = (x, y)^T$. The conditional mean and conditional variance of random variables are estimated by the simple kriging method (Deutsch and Journel, 1998; Sun et al., 2016; Zhang, 2005):

$$\langle I(x) \rangle^{(c)} = \langle I(x) \rangle + \sum_{i=1}^n \mu_i [I(x_i) - \langle I(x_i) \rangle] \quad (7)$$

$$Var^{(c)}(x) = C_I(x, x) - \sum_{i=1}^n \mu_i C_I(x, x_i) \quad (8)$$

where $\langle I(x) \rangle^{(c)}$ and $Var^{(c)}(x)$ are the conditional mean function and the conditional variance function, respectively; $\langle I(x) \rangle$ and $C_I(x_1, x_2)$ are the unconditional mean and the unconditional covariance function, respectively; and since the virtual observations $I(x_i)$ were equal to the unconditional mean $\langle I(x) \rangle$, the conditional mean and the non-conditional mean had the same distribution. The random perturbation values at the center of the plants were equal to the virtual observations. The virtual observations changed the covariance and variance distribution of the conditional stationary field, and reduced the variance within the correlation length range. The functions μ_i are weighting functions representing the relative importance of each 'virtual observations' $I(x_i)$ in predicting the value of $\langle I(x) \rangle^{(c)}$ at location x , and can be solved from the following kriging equations:

$$\sum_{i=1}^n \mu_i C_I(x_i, x_j) = C_I(x, x_j) \quad j = 1, 2, \dots, n \quad (9)$$

Maize is an important representative crop in the Loess Plateau and is widely cultivated in Ningxia. The root systems of maize is evenly distributed in the soil, with a symmetrical left-right distribution. Therefore, it was selected as the object of study in this paper to study the effect of plant roots on desiccation cracking. The non-stationary random

field parameters and their physical meanings are shown in Table 1. Each parameter can be varied to produce different random fields. Notably, the lateral distribution radius of RDDF r_m and the maximum critical strain value h_c act together to reflect the strengthening range and strengthening the strength of the root systems to the soil. By adjusting them, it is possible to describe the influence of different plant roots on soil reinforcement. Simulation of desiccation cracking behavior of different soil types needs to be achieved by adjusting the mean value, variance, and correlation length.

3. Materials and methods

3.1. Field experiments details

The experiments were performed at the Efficient Use of Water Resources in Arid Modern Agriculture laboratory at the Ministry of Educational Engineering Research Center, located in the western Ningxia province, China. The experimental area ($37^\circ 48' N$, $105^\circ 70' E$), located in Zhongning County, Ningxia, China, is in the transition zone between the Inner Mongolia Plateau and the Loess Plateau. With a climate of north temperate monsoon climate the average annual temperature in this area is $9.5^\circ C$, the average annual precipitation is 202.1 mm, and the average annual potential ET is 1947.1 mm.

Maize has been grown in the experimental area for a long time and the cropping system is one crop per year. Sowing started in mid to late April and harvesting took place mid to late September. The land in the experimental area was plowed once a year before sowing, and irrigated in stages during sowing and growth, using flood irrigation. The irrigation water source was taken from the Yellow River, with an average annual total irrigation water volume of 4350 mm. The experimental soil was silty clay according to the USDA soil classification, and its physical properties are shown in Table 2.

The experiment was conducted in the field in September 2018. Eight $100 \text{ cm} \times 100 \text{ cm}$ square areas, numbered S1 to S8, were set up in the experimental area, where zones S1 to S7 had natural maize roots, and zone S8 was used to determine soil moisture content. The experimental layout is given in Fig. 1. Before the experiment, the stems and leaves of the corn were removed, and the soil of the upper 1 cm layer of the experimental area was loosened and removed to level the soil surface while keeping the root structure undisturbed. Then, the ponded water was applied to the soil surface and a constant rate of water injection was maintained. A TDR (Time Domain Reflectometry) was applied to measure soil water content to ensure that the soil was fully saturated in the upper 10 cm. Once saturated, the soil surface swelled and was free of cracks. The experiment was carried out over three consecutive days from 8:00–18:00 with a Canon EOS 5DII digital camera taking images of crack formation and development every two hours. To avoid the influence of natural rainfall on the cracks in the experiment area, insulation and rainproof materials were adopted to cover the experimental area during the night. There was no rainy weather during the experimental period.

3.2. Photography and image processing

Digital images needed to be processed to further analyze the geometric characteristics of the cracks. The typical procedure for digital image processing consists of the following four steps (Zhu et al., 2020),

Table 1
List of Parameters of non-stationary Random Field.

Random fields parameters	Physical meaning
Mean value of critical strain $\bar{e}$	The elasticity of the material
Variance δ^2	Microscopic heterogeneity of soil
Correlation length λ (cm)	The spatial correlation of soil properties
Maximum critical strain value h_c	The critical strain of the soil at the location of the plant centers
The number of maximum critical strain locations n	The number of plants
The lateral distribution radius of RDDF r_m (cm)	The influence range of root reinforcement

Table 2
Physical soil properties in the experimental area.

Soil type	Depth (cm)	Particle size distribution (%)			Bulk density ($\text{g}\cdot\text{cm}^{-3}$)	Porosity (%)
		> 50 μm	2–50 μm	< 2 μm		
Silty clay	0–10	3.7	51.2	45.1	1.46	45.4
	10–20	4.3	50.4	45.3	1.47	45.1
	20–30	4.4	50.1	45.5	1.53	44.2
	30–50	4.5	50.3	45.2	1.52	44.7

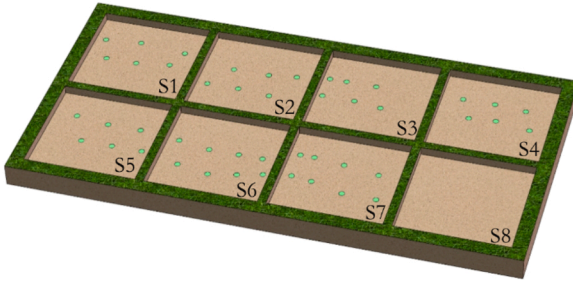


Fig. 1. Experiment layout.

cropping the non-experimental part of the image, processing the image size to 2048×2048 pixels and applying the image geometrical corrections, removing the roots from cracked images with roots, converting the color image of the crack patterns into a gray image and after converting it into a binary image using a threshold division method, the image was processed through noise removal (Fig. 2). The black lines and white areas in the binary image represent crack networks and aggregates, respectively.

3.3. Morphological measures

In this paper, the Minkowski number M_k was used to describe the geometric characteristics of the crack networks. For the two-dimensional crack pattern on the soil surface, the Minkowski number M_k is expressed by the three fundamental properties of the crack area A , length L , and Euler number E . The calculation formula is as follows (Vogel et al., 2005a; Vogel and Weller, 2010; Zhu et al., 2020):

$$M_0(X) = A(X)[L^2] \quad (10)$$

$$M_1(X) = \int_{\delta X} ds = L(X)[L] \quad (11)$$

$$M_2(X) = \int_{\delta X} \frac{1}{r} ds = 2\pi E(X) \left[- \right] \quad (12)$$

where X is the crack network structure. δX denotes the boundary of the structure X . ds represents the boundary element and r is the radius of curvature of the boundary. M_0 is the area of the cracks measured by the number of black pixels in the binary images, M_1 is the boundary length of the crack and M_2 represents a topological measure describing the connectivity of the crack structure by the Euler number $E(X)$, which is defined as the number of isolated objects N (closed convex) minus the number of holes C (closed concave), i.e., $E = N - C$.

To compare the crack structure from images with different sizes, Minkowski densities m_k were introduced by dividing Minkowski numbers M_k with the area of the image Δ (or $(2\pi\Delta)$):

$$m_k(X) = \frac{M_k}{\Delta} k = 0, 1 \quad (13)$$

$$m_2 = M_2 / (2\pi\Delta) \quad (14)$$

where Δ is the size of the image and m_2 is applied to describe the density of the Euler number (Vogel et al., 2005a; Zhu et al., 2020).

3.4. Model evaluation method

The performance of the model was evaluated by comparing the Minkowski density m_k of the experimental and simulated crack. In this paper, four evaluation indicators of determination coefficient R^2 , consistency index IA , $BIAS$, and root mean square error $RMSE$ were adopted to quantify the difference between the experimental and simulated Minkowski densities. The mathematical expressions for R^2 , IA , and $BIAS$ are as follows (Legates and McCabe, 1999; Moriasi et al., 2007; Zhu et al., 2020):

$$R^2 = 1 - \frac{\sum_{i=1}^N (S_{wos,i} - S_{wsm,i})^2}{\sum_{i=1}^N (S_{wos,i} - \bar{S}_{wos})^2} \quad (15)$$

$$IA = 1 - \frac{SE}{\sum_{i=1}^N [|S_{wsm,i} - \bar{S}_{wos}| + |S_{wos,i} - \bar{S}_{wos}|]^2} \quad (16)$$

$$BIAS = \frac{\sum_{i=1}^N |S_{wos,i} - S_{wsm,i}|}{\sum_{i=1}^N S_{wos,i}} \quad (17)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (S_{wos,i} - S_{wsm,i})^2}{N - 1}} \quad (18)$$

where $S_{wsm,i}$, $S_{wos,i}$ denote the simulated and observed values, respectively, $\bar{S}_{wos}$ represents the average values of $S_{wos,i}$ and N is the number of samples. The optimal value of R^2 and IA is 1, and the optimal value of $BIAS$ and $RMSE$ is 0.

4. Results and discussion

Zone S1 was chosen for model calibration and zones S2 to S7 were selected to validate the model. 400×400 nodes represented the $100 \text{ cm} \times 100 \text{ cm}$ area. The Minkowski densities of the experimental and simulated images were calculated and compared, and the results of the model calibration and verification were evaluated through the corresponding evaluation indexes.

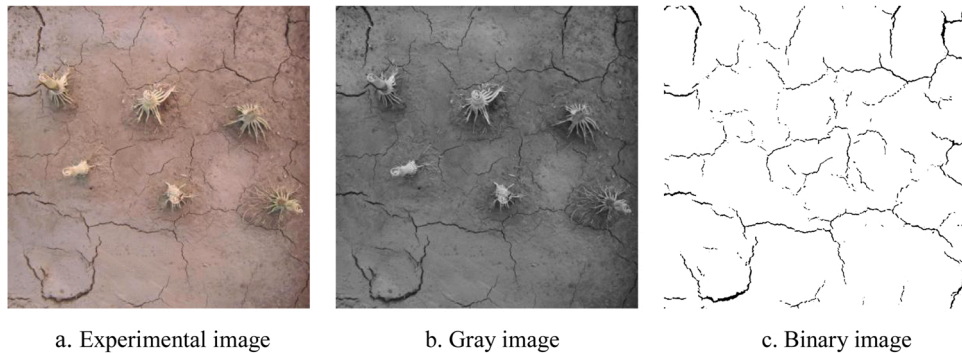


Fig. 2. Image binarization process.

4.1. Calibration of the model

The objective of the model calibration was to capture the overall crack pattern observed in the field by adjusting the input parameters. The Monte-Carlo method was applied for parameter calibration in this study, using twenty-four cores (Intel Xeon Gold 6128 CPUs running at 3.4 GHz with 128 GB of RAM) on workstations, generating a total of 1618,496 realizations. The parameter combination calculated by the Monte-Carlo method is shown in Table 3.

The concept of time was not addressed in the model of Vogel et al. (2005b), but the authors interpreted the spring tension $u=1-d$ as a function of time or soil moisture content. Zhu et al. (2020) calibrated the model time by minimizing the experimental and simulated Minkowski densities. In this study, the method of Zhu et al. (2020) was applied to match the model time through four-time points of the experiment (2 h, 10 h, 18 h and 26 h), with each corresponding to a certain spring relaxation length d in the model.

P_{mi} and P were defined according to three evaluation indicators (R^2 , IA , $BIAS$) with the following equations (Zhu et al., 2020):

$$P_{mi} = R^2 + IA - BIAS \quad i = 0, 1, 2 \quad (19)$$

$$P = \sum_{i=0}^2 P_{mi} \quad (20)$$

P_{mi} ($i = 0, 1, 2$) represents the comprehensive evaluation index of area, length, Euler number density respectively, and P represents the overall evaluation index. $P_{mi} \leq 2$ and its optimal value is 2; $P \leq 6$ and its optimal value is 6. The objective of calibration is to find an optimized parameter set that can maximize the value of P .

Based on the results of the Monte-Carlo simulation, the P value of each parameter combination was calculated, and the optimal parameters of the model were finally determined (see Table 4). The evaluation index $RMSE$ was also calculated as a reference for the rationality of the parameters. The parameter set shown in Table 4 achieved the maximum P value of 5.7128 (see Table 5) among all parameter combinations of the Monte-Carlo simulation, which ensured the optimality of the model parameters. Fig. 3 shows the critical strain random field at calibrated parameters and the comparison between the experimental crack and the simulated crack image is demonstrated in Fig. 4. It can be observed that the simulated and experimental crack images present approximately the same morphological characteristics, that is, the soil is less cracked close to the root systems and more significantly cracked at locations away from the root systems. The matching of the Minkowski densities of the simulated and experimental crack images is shown in Fig. 5. The Minkowski density development rule of simulated cracks and experimental cracks is almost identical.

4.2. Validation of the model

The model parameters were validated by the experimental crack in zones S2 to S7. The purpose of model validation was to evaluate the accuracy of the model in simulating experimental cracks in other zones under optimal parameter conditions. The location and number of plants

Table 4

Calibrated model parameters for this study.

Model parameters	Calibration value in this study
Mean value of critical strain $\bar{\epsilon}$	0.1
Variance δ^2	0.36
Correlation length λ (cm)	7.5
The number of the maximum critical strain locations n	The number of plants
Maximum critical strain value h_c	0.4
Relaxation n_{it}	20
Friction μ	0.0015
Shrink minimum length of nodes d_{min}	0.9
The lateral distribution radius of RDDF r_m (cm)	8

Table 5

Calibration statistics for the model area density, length density and Euler number density.

Statistical indicators	Area density m_0	Length density m_1	Euler number density m_2
$R^2[-]$	0.970	0.917	0.987
$IA[-]$	0.993	0.973	0.997
$BIAS[-]$	0.0663	0.0318	0.0261
$RMSE$	0.0002	0.0128	0.0249
P_{mi}	1.8967	1.8589	1.9579
P	5.7128		

Note: R^2 , IA and $BIAS$ are dimensionless; $RMSE$ has same as the dimension of the statistical variable.

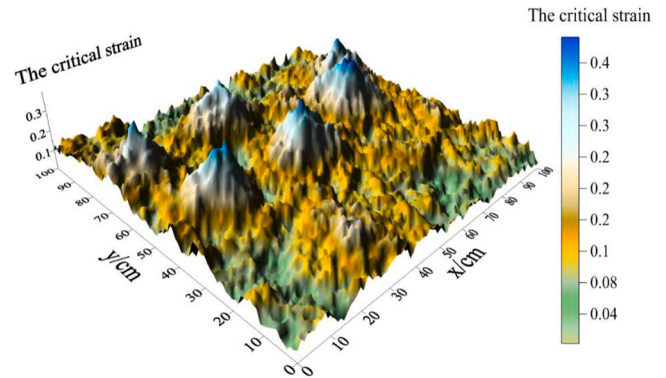


Fig. 3. Critical strain random field realization at calibrated parameters (S1).

in the experimental zones were different. There were six corn roots in zones S2 to S5 and eight corn roots in zones S6 to S7. Each group of experiments was validated separately by varying the number of random seeds, namely $nseed$, to generate 100 random realizations under the optimal input parameters, and then 100 sets of crack images that conformed to the optimal parameters were then obtained. The difference between the Minkowski density of each group of experimental cracks and the average value of the corresponding 100 groups of simulated cracks Minkowski density was compared respectively, and four evaluation indexes of R^2 , IA , $BIAS$, and $RMSE$ were used for evaluation. The evaluation indicators validated by the six experiments are presented in Table 6. Among the average evaluation indexes of area, length, and Euler number density, R^2 and IA were all greater than 0.8, and $BIAS$ and $RMSE$ were both less than 0.11. Fig. 6 displays a comparison between the simulated and measured mean values during crack formation, and the standard deviation was calculated. Due to the differences in the number of plant roots in the experiment zones, four six-root experimental zones and two eight-root experimental zones were compared, respectively. The model verification results are favorable, and can simulate the development process of soil desiccation cracking under the

Table 3

Parameter ranges and increments of Monte-Carlo method.

Model parameters	range	increment
Mean value of critical strain $\bar{\epsilon}$	0.05–0.2	0.05
Variance δ^2	0.2–0.4	0.02
Correlation length λ (cm)	1–10	0.5
Maximum critical strain value h_c	0–1	0.1
Relaxation n_{it}	10–25	1
Friction μ	0.001–0.002	0.0001
Shrink minimum length of nodes d_{min}	0.7–0.999	0.001
The lateral distribution radius of RDDF r_m (cm)	1–20	0.5

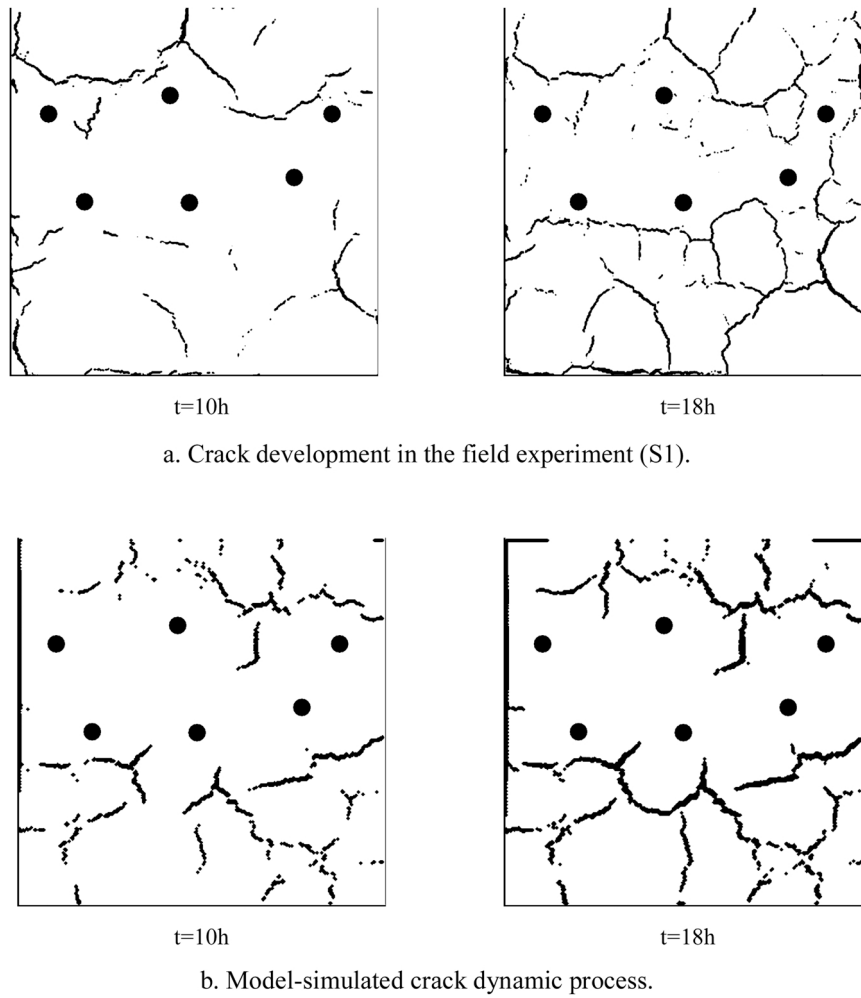


Fig. 4. Comparison of simulated and experimental crack images. The black dot indicates the position of the plant root.

influence of root systems in different experimental zones.

5. Sensitivity analysis

The simulation results of the desiccation cracking model of farmland soils under the influence of the root systems were jointly controlled by mean critical strain $\bar{\epsilon}$, variance δ^2 , correlation length λ , relaxation parameter n_{it} , and friction μ , maximum critical strain value h_c , the lateral distribution radius of RDDF r_m , the number of maximum critical strain locations n . Vogel et al. (2005b) analyzed the sensitivity of the four parameters of the model mean, variance, correlation length, and friction. Zhu et al. (2020) introduced the correlation length into the Vogel et al. (2005b) model for an unconditional simulation of soil desiccation cracking, and analyzed the influence of the correlation length λ . However, the effect of a single parameter and each parameter combination on the area density, length density, and Euler number density index has not been quantitatively analyzed. Therefore, the Sobol global sensitivity analysis method was introduced to quantitatively study the sensitivity of single parameter and parameter combination of soil cracking model under the influence of root systems to objective functions (Minkowski densities) in Monte Carlo simulation.

5.1. Sobol global sensitivity analysis method

In order to quantitatively evaluate the degree of influence of each input parameter and inter-parameter interaction on the output variable, the Sobol global sensitivity analysis was introduced to decomposing the

variance of the output variable. Assume that y is the model output variable, X_i is the model input parameters (m in total). Variance $D(y)$ can be decomposed as:

$$D(y) = \sum_i D_i + \sum_{i < j} D_{ij} + \sum_{i < j, k} D_{ijk} + D_{1,2,\dots,m} \quad (21)$$

Where D_i represents the variance of parameter i , D_{ij} represents the variance of the interaction of parameters x_i and x_j .

The sensitivity of a single parameter i can be reflected by the first-order sensitivity index S_i . The total order sensitivity index S_{Ti} indicates the joint influence of a single parameter with all other parameters and the formula was as follows:

$$S_i = \frac{D_i}{D} \quad (22)$$

$$S_{Ti} = 1 - \frac{D_{\sim i}}{D} \quad (23)$$

Where $D_{\sim i}$ represents the variance of other parameters except parameter x_i .

5.2. Sensitivity analysis of model input parameters

The sensitivity of most input parameters (mean critical strain $\bar{\epsilon}$, variance δ^2 , isotropic correlation length λ , relaxation parameter n_{it} , and friction μ , maximum critical strain value h_c , the lateral distribution radius of RDDF r_m) in Monte Carlo simulation were analyzed. Since the

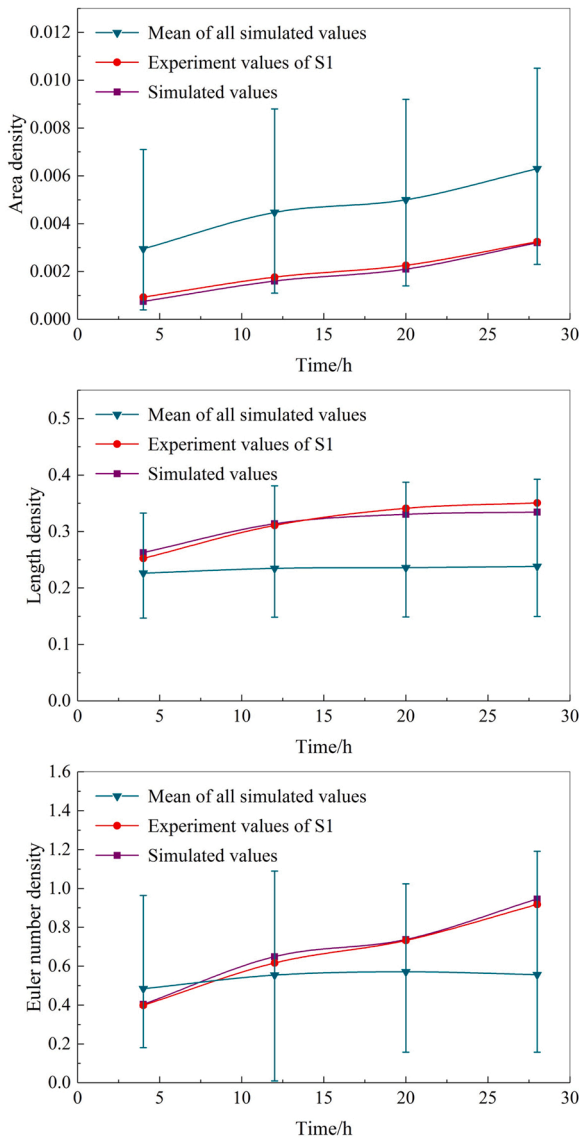


Fig. 5. Match between simulated and measured values in S1 after calibration (red, pink) and average of all Monte Carlo simulations (green) with vertical bars showing the total range of simulated values.

number of plants in the experiment was determined, its sensitivity was not analyzed. The maximum critical strain value h_c represented the strength of the strengthening effect of the root systems on the soil. The lateral distribution radius of RDDF r_m represented the influence range of root reinforcement.

Fig. 7 displays the first-order and total order sensitivity indices of the seven input parameters to Minkowski density. The first-order sensitivity index is classified as a sensitive parameter when it exceeds the threshold value of 0.1. As shown in Fig. 7, taking area density, length density and Euler number density as objective functions, the sensitivity indices of mean critical strain $\bar{\epsilon}$, variance δ^2 , friction f , and the lateral distribution radius of RDDF r_m were all more than 0.1. Besides, the area density and length density were sensitive to the maximum critical strain value h_c , and the Euler number density was sensitive to the correlation length λ . These parameters played a decisive role in the crack pattern of the experimental area and were the key parameters for the model to simulate soil shrinkage cracking. Area density was most sensitive to r_m with a sensitivity index of 0.56, which was due to the increase of r_m expanded the influence range of the maximum critical strain value h_c and increased the area of the high critical strain region, resulting in the

Table 6

Validation statistics for area density, length density, and Euler number density.

Statistical indicators		R^2	IA	BIAS	RMSE
S2	Area density m_0	0.942	0.986	0.0735	0.0002
	Length density m_1	0.838	0.944	0.0489	0.0189
	Euler number density m_2	0.948	0.987	0.0392	0.0460
S3	Area density m_0	0.936	0.986	0.0979	0.0002
	Length density m_1	0.801	0.918	0.0568	0.0255
	Euler number density m_2	0.886	0.963	0.0982	0.0986
S4	Area density m_0	0.912	0.975	0.1200	0.0003
	Length density m_1	0.829	0.933	0.0444	0.0227
	Euler number density m_2	0.756	0.947	0.1070	0.0965
S5	Area density m_0	0.836	0.958	0.1390	0.0004
	Length density m_1	0.783	0.928	0.0495	0.0224
	Euler number density m_2	0.848	0.969	0.0724	0.0696
S6	Area density m_0	0.890	0.973	0.1180	0.0003
	Length density m_1	0.833	0.935	0.0520	0.0638
	Euler number density m_2	0.865	0.971	0.0590	0.0219
S7	Area density m_0	0.922	0.979	0.0843	0.0003
	Length density m_1	0.725	0.908	0.0515	0.0219
	Euler number density m_2	0.749	0.930	0.1340	0.1209
Average for S2-S7	Area density m_0	0.906	0.976	0.1055	0.0003
	Length density m_1	0.802	0.928	0.0505	0.0292
	Euler number density m_2	0.842	0.961	0.0850	0.0755

Note: R^2 , IA and BIAS are dimensionless; RMSE has same as the dimension of the statistical variable.

increase of the crack-free region in the simulated crack image and reduced the overall area of the soil crack. Length density was most sensitive to variance with a sensitivity index of 0.67, which was due to the increase of variance δ^2 enhanced the heterogeneity of the critical strain random field, leading to the increase of the number of isolated cracks in soil and the increase of the overall length of cracks. Euler number density was most sensitive to correlation length λ with a sensitivity index of 0.56, which was due to the increase of correlation length λ reduced the local heterogeneity of the critical strain random field, promoting the interconnection of isolated crack and forming a connectivity crack network.

From the perspective of the total order sensitivity indices, when the parameters interact, the total order sensitivity indices of input parameters for different objective functions (area density, length density and Euler number density) were between 0.61 and 0.81, 0.53–0.93 and 0.51–0.74, respectively, and the differences were not significant. For each sensitivity parameter, the proportion of first-order sensitivity index was smaller than that of total order sensitivity index. It indicated that compared with the single action of parameters, the interaction of parameters affected the model results to a greater extent.

6. Conclusion

In this paper, the non-stationary random field was introduced into the Vogel et al. (2005b) model to develop a model of desiccation cracking of farmland soil influenced by root systems, and the parameters in the model were determined according to the experimental results. And lastly, a sensitivity analysis was carried out on the parameters of the model. The main conclusions are presented as follows.

1) The model developed in this paper has performed well in terms of calibration and verification. The model can reproduce the dynamic morphological characteristics of the desiccation cracking network of farmland soil influenced by root systems in nature, indicating that the model is reliable and practical.

2) The model reproduces the complex phenomenon of the development of crack patterns influenced by root systems. It makes full use of

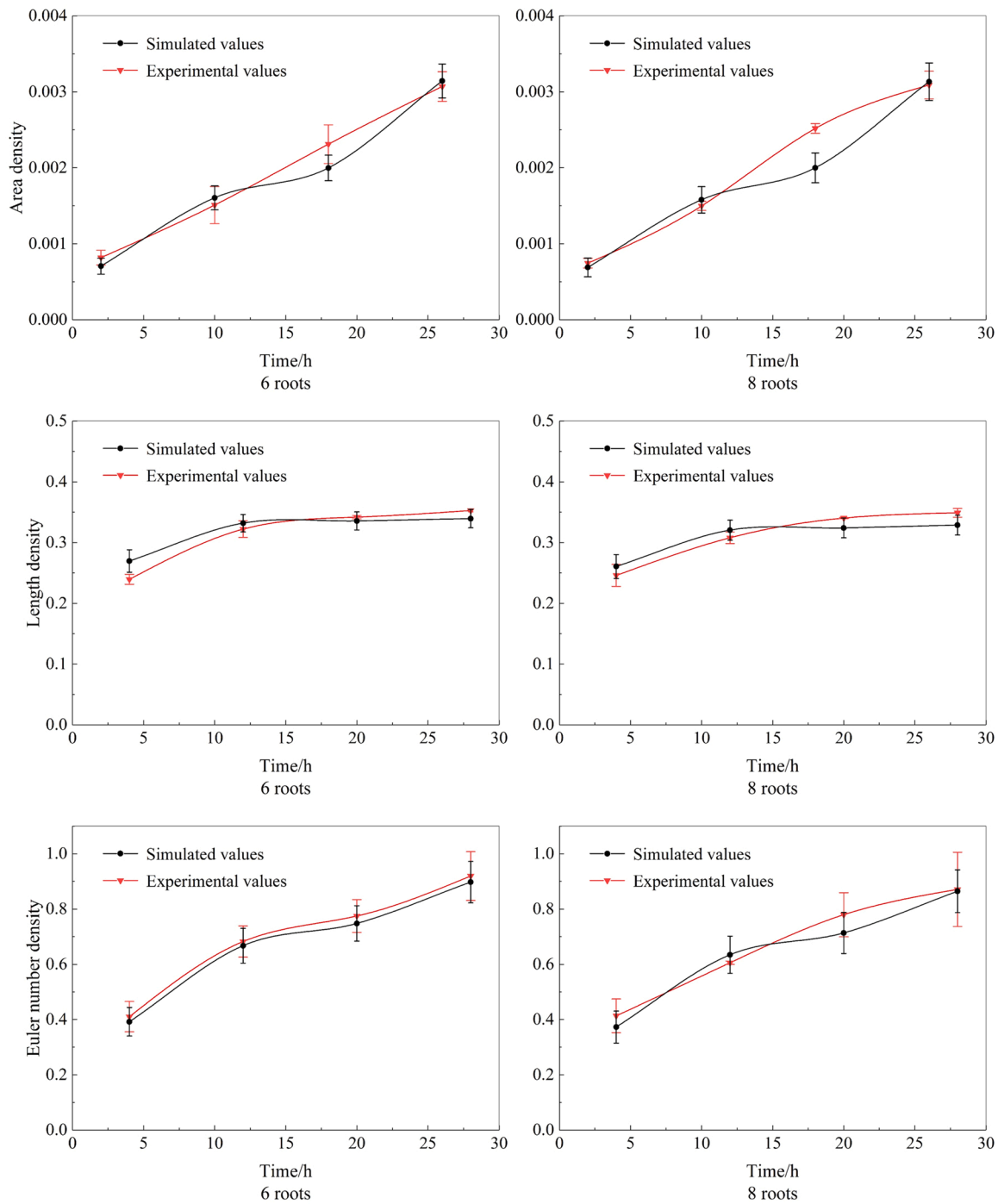


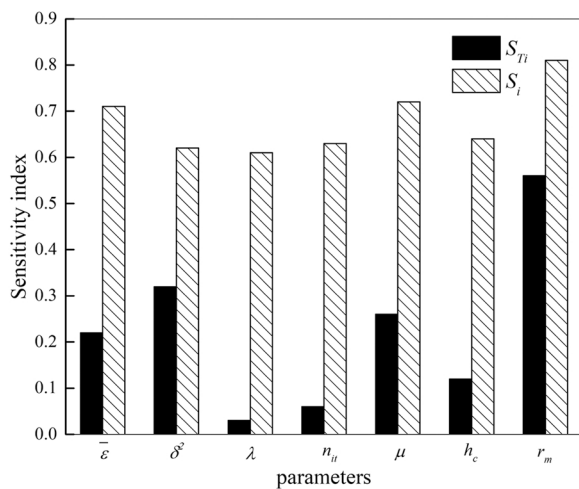
Fig. 6. Comparison between simulated and measured values in the validation of the model. The error bar represents the standard deviations of the simulated and experimental value, respectively.

the condition simulation technique and the root density function to describe the spatial variation structure of the soil critical strain. The calculation results fully reflect the strengthening effect of root systems on the soil and the restriction of root systems on desiccation cracking, and illustrates the rationality of non-stationary random fields to describe the critical strain of farmland soil influenced by root systems.

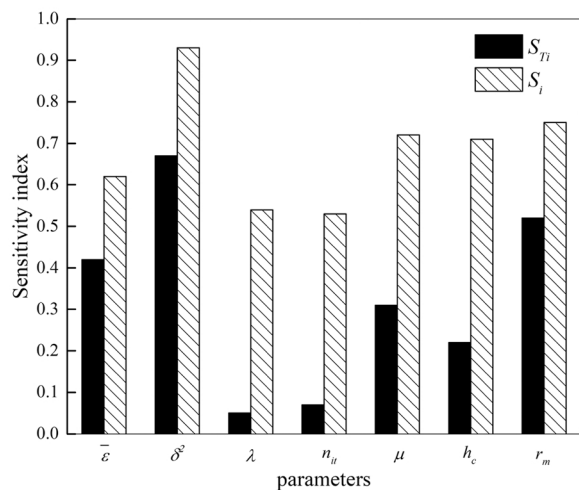
3) For input parameters in Monte-Carlo simulation, the mean critical strain $\bar{\epsilon}$, variance δ^2 , friction f , and the lateral distribution radius of RDDF r_m showed significant sensitivity to the objective functions (area density, length density, and Euler density). These parameters played a decisive role in the simulated crack pattern and were the key parameters for the model to simulate soil shrinkage cracking. The area density was most

sensitive to the r_m . The length density was most sensitive to the variance δ^2 . Euler number density was most sensitive to the correlation length λ . Compared with the single action of parameters, the interaction of parameters affected the model results to a greater extent.

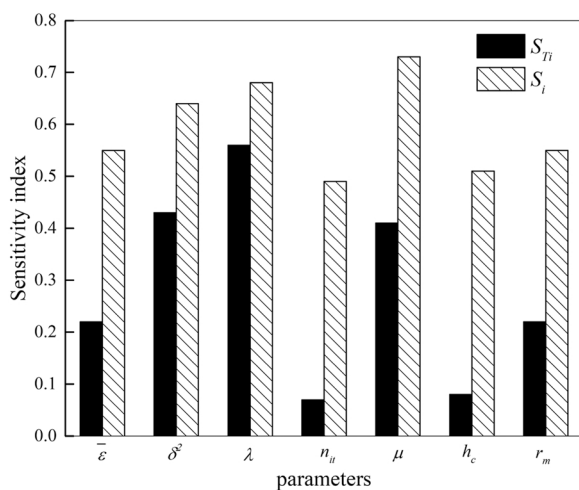
By spring shrinkage, this model described the phenomenon of soil shrinkage caused by water evaporation, but did not consider changes in soil water content. In reality, the degree and rate of water content change can significantly affect the crack development rate. In the subsequent study, an attempt can be made to couple the soil shrinkage cracking model with the real water content of the soil to establish the relationship between water content and crack evolution, so as to improve the prediction accuracy and reliability of the model.



(a) Area density



(b) Length density



(c) Euler number density

Fig. 7. First order and total order sensitivity of model parameters for different Minkowski density.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data Availability

No data was used for the research described in the article.

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